MASSIMILIANO REMIGIO

Chemist, PhD · Analytical and Materials Chemistry · R&D, QC, and Process

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PROFESSIONAL PROFILE

Chemist completing a joint PhD at KU Leuven and Université libre de Bruxelles, with more than three years of daily hands-on lab work and earlier experience in industry. I bring analytical chemistry (HPLC-MS method development, GC-MS, NMR, ICP-MS at ppb to ppt levels, method validation), materials and surface characterisation (XPS/UPS, SEM-EDX, XRPD, Raman, FT-IR, AFM), and process experience from operating a pyrolysis pilot plant from commissioning to a robust, repeatable process. I plan experiments by multivariate DOE, analyse data in Python, and document to GLP-comparable standards with SOPs, deviation logs, and CAPA records. I am looking for a role in R&D, analytical or quality control, or process development.

TECHNICAL EXPERTISE

Analytical chemistry

- HPLC-MS · purity method development
- NMR · routine and quantitative
- Method validation · LOD/LOQ
- GC-MS · residual solvents
- ICP-MS/OES · ppb to ppt trace work
- Ion chromatography, AAS, titration

Materials and surface characterisation

- XPS/UPS (UHV) · surface chemistry
- XRPD, Raman, FT-IR, AFM-IR
- DSC/TGA and BET
- SEM-EDX · morphology, composition
- AFM · roughness, conduction
- Electrochemistry · EIS, CV

Process, quality, and data

- Pilot plant · commissioning to output
- SOPs, deviation logs, CAPA records
- DOE · multivariate experiment design
- Python · statistics, data processing

PROFESSIONAL EXPERIENCE

PhD Researcher, Analytical and Materials Chemistry

KU Leuven / Université libre de Bruxelles · *2022 to Present* · Leuven, Belgium

Joint doctorate on molecular self-assembly, thin films, and chiral surface networks, run on a multi-instrument platform that I operate myself every day.

- **Analytical methods:** Developed HPLC-MS methods for purity determination and validated methods for linearity, precision, accuracy, and LOD/LOQ.
- **Surface and materials analysis:** Daily XPS/UPS under UHV, SEM-EDX, XRPD, Raman, FT-IR, and AFM, including AFM-IR to verify surface functionalisation.
- **Thin films and devices:** Prepared films by CVD, PVD, and solution deposition, did cleanroom work and etching to produce my own substrates, and worked on viologen electrochromic devices.
- **Experiments and data:** Planned multivariate DOE campaigns and built Python models linking process parameters and film structure to measured properties.
- **Quality:** Set up, calibrated, and verified instruments daily, and kept SOPs, deviation logs, and CAPA records to GLP-comparable standards.
- **Teamwork:** Ran three workstreams in parallel and coordinated campaigns with academic and industrial partners in Belgium, Germany, and Italy.

Publications: *Nature Materials (submitted)* · *J. Am. Chem. Soc. 2025* · *Cryst. Growth Des. 2025.*

MSc Researcher, Pilot-Scale Process Development

University of Bologna · *2019 to 2021* · Bologna, Italy

- **Lab to pilot scale:** Operated a pyrolysis pilot plant from commissioning through sustained production, defined operating protocols, and troubleshot instabilities until the process was robust and repeatable.

- **Process to product:** Correlated process conditions with product properties by Raman, FT-IR, XPS, TGA, BET, and SEM-EDX, refining the recipe at increasing scale.

Industrial QC and Materials Intern

Walter Tosto S.p.A. · *2017 to 2020* · Ortona, Italy

- **Specifications:** Qualified components against technical specifications in a certified production environment, with formal, traceable documentation.
- **Materials and corrosion:** Studied corrosion and degradation of high-pressure process materials by potentiodynamic polarisation, EIS, and cyclic voltammetry; investigated failures by SEM-EDX and XRPD.

Radiochemical Analyst, Nuclear Science Internship

Borexino Neutrino Experiment, INFN · *2016* · Gran Sasso, Italy

- **Trace analysis:** ICP-MS at ppb to ppt levels with method development, validation, and strict contamination control.

Analytical Chemistry and QA Intern

La Valle's Chemistry Laboratory · *2015 to 2016* · Italy

- **Testing against specifications:** Analysed water, food, and waste samples by HPLC, ion chromatography, GC, AAS, and MS against defined specifications; investigated OOS results.
- **Pharmacopoeial methods:** Tested solutions, creams, and emulsions to prescribed pharmacopoeial and customer methods with full traceability.

QUALITY FRAMEWORK

GMP/GLP awareness · ICH Q2(R2) method validation · ALCOA+ data integrity · CAPA and deviation management · instrument lifecycle management

EDUCATION

PhD, Chemistry (Surface Science and Materials) KU Leuven and ULB · 2022 to Present

MSc, Chemistry (Organic and Green Chemistry) University of Bologna · 2019 to 2021
Thesis: mechanochemical, solvent-free peptide synthesis

BSc, Chemistry (Materials and Chemistry) University of Bologna · 2016 to 2019
Thesis: retro-inverso peptide synthesis

LANGUAGES

Italian Native · **English** Fluent · **Dutch** Actively developing, resident in Leuven · **French** Basic

Analytical chemistry · materials characterisation · process development · DOE · Python · Leuven based